CSC 675/775

Introduction to Database Systems

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What are databases and why are they important?

IEEE Top Programming Languages

- Graphs from googling “popular programming languages”
  - IEEE stands for “Institute of Electrical and Electronics Engineers”
- The “Spectrum” ranking is weighted towards the profile of the typical IEEE member
- The “Trending” ranking seeks to spot languages that are in the zeitgeist
- The “Jobs” ranking measures what employers are looking for.

Programming Languages by Spectrum

IEEE Spectrum | The Top Programming Languages 2024

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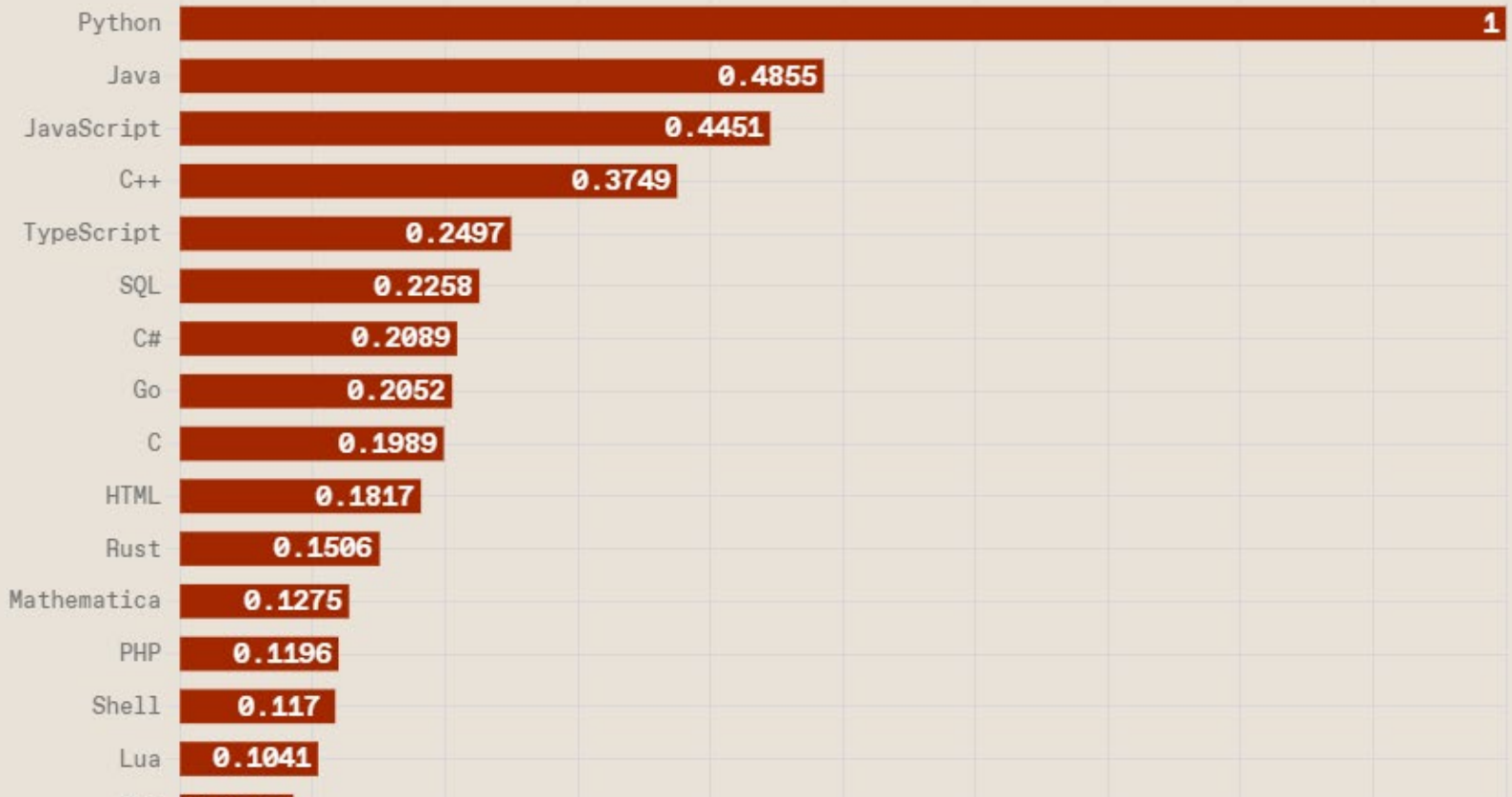
PYTHON | SQL | RUST

TYPESCRIPT

Top Programming Languages 2024

Click a button to see a differently weighted ranking

Spectrum | Trending | Jobs



Programming Languages by Trending

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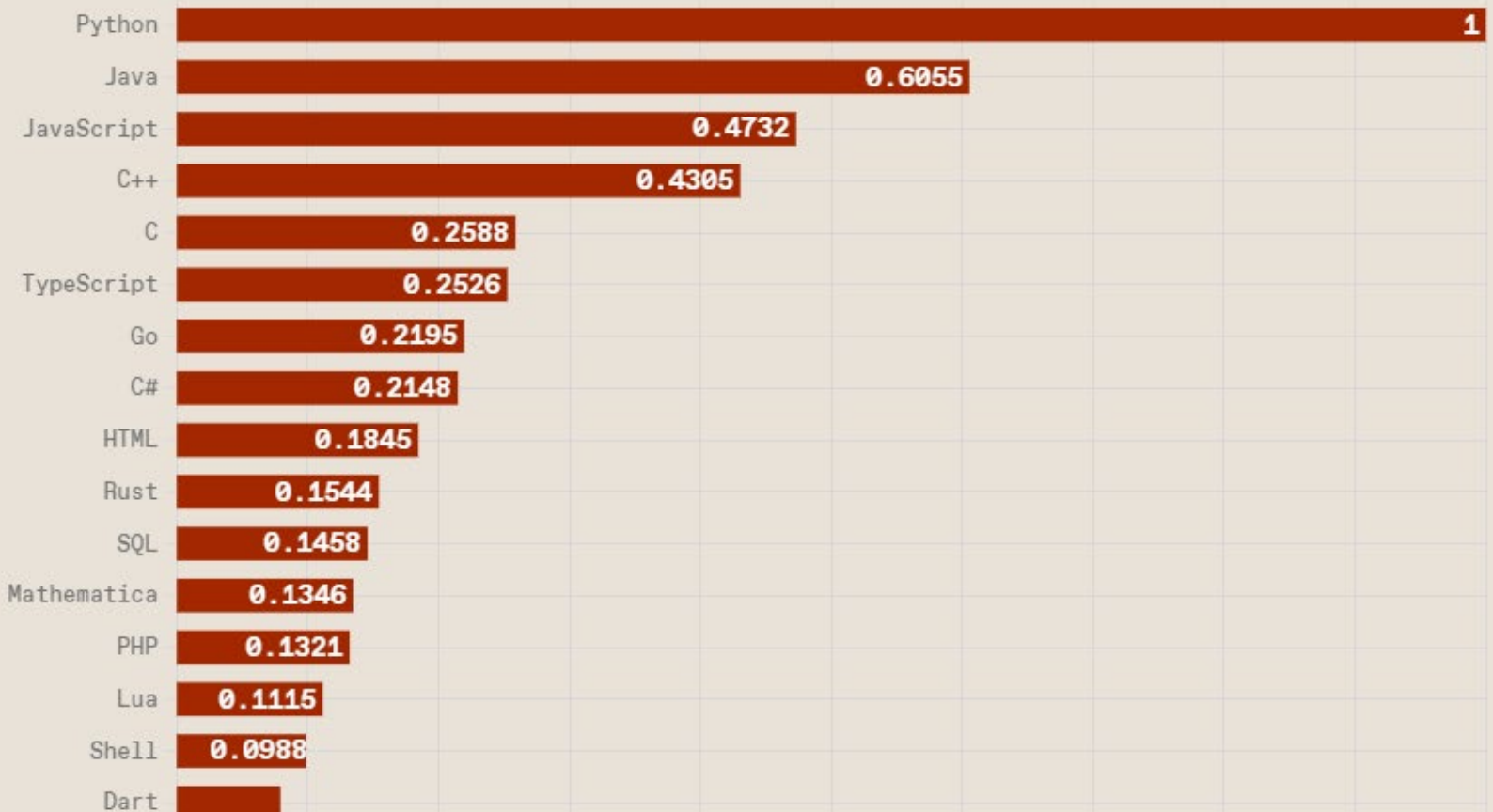

Programming Languages by Jobs

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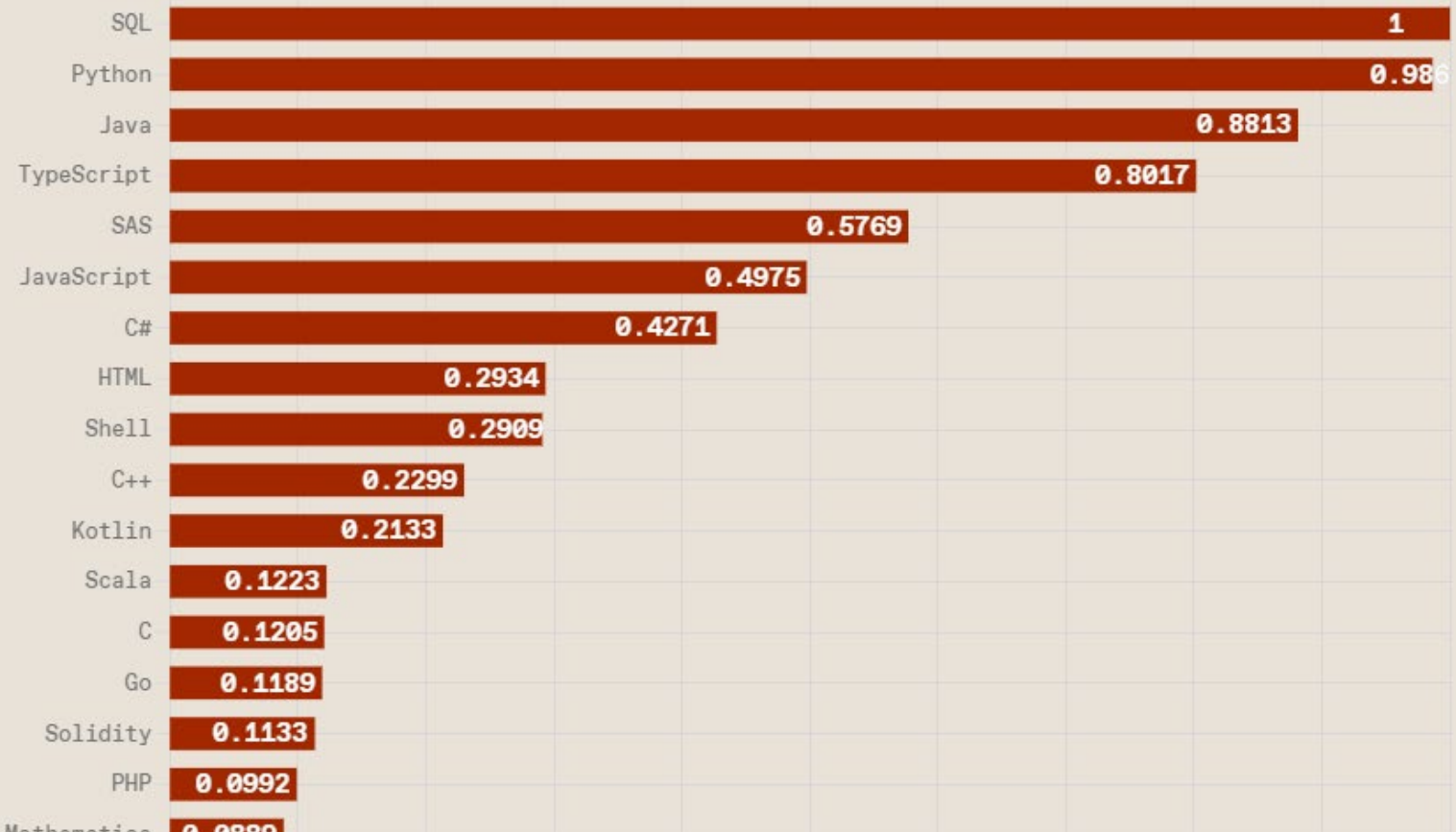


Database Management System (DBMS)

- Storage and retrieval of data
  - Manage collections of interrelated structured data
  - Multiuser access
- Transaction Management
- Recovery in case of crash

Relational Model

- Collection of tables describing data
  - Everything is stored in tables (rows and columns like spreadsheets)
    - Relations, Relationships, semantics, constraints
  - Each table has columns defined
    - Structured data
    - Datatypes predefined and apply to all data in the column
  - Rows are instances of data stored in the table

- Conceptual model
  - Entity Relationship Diagram (ERD)
    - Visual description of tables in a database

Relational Model

ID	name	dept_name	budget
39827	John	Physics	85000
48836	Kim	Comp. Sci	50000
29987	Califieri	Biology	90000
56655	Katz	Physics	85000

Relational Model

ID	name	dept_id
398227	John	1
48836	Kim	2
29987	Califeri	3
56655	Katz	1

(a) Instructor table

dep_id	dept_name	budget
1	Physics	85000
2	Comp. Sci.	50000
3	Biology	90000

(b) Department table

Data Definition Language - DDL

```sql
CREATE TABLE INSTRUCTOR (
   ID   INT                NOT NULL,
   NAME VARCHAR (20)       NOT NULL,
   DEPARTMENT  INT         NOT NULL,
   PRIMARY KEY (ID)
   FOREIGN KEY (DEPT_ID) REFERENCES DEPARTMENT(DEPT_ID)
);

CREATE TABLE DEPARTMENT (
   DEPT_ID   INT                 NOT NULL,
   NAME VARCHAR (20)       NOT NULL,
   BUDGET  INT        NOT NULL,
   PRIMARY KEY (DEPT_ID)

);
```

Example DDL definition of tables instructor and department (MySQL)

Data Manipulation Language - DML

- Query language
  - Complete
    - Selection, Projection, Cartesian product, Union, Intersection, Set difference
  - Industry standard is Structured Query Language (SQL)
    - SQL is the most widely used data access language
    - SQL queries, Triggers, Procedures, Functions

Data Manipulation Language - DML

```
SELECT * FROM INSTRUCTOR WHERE DEPT_ID = 1;
```

ID	name	dept_id
398227	John	1
56655	Katz	1

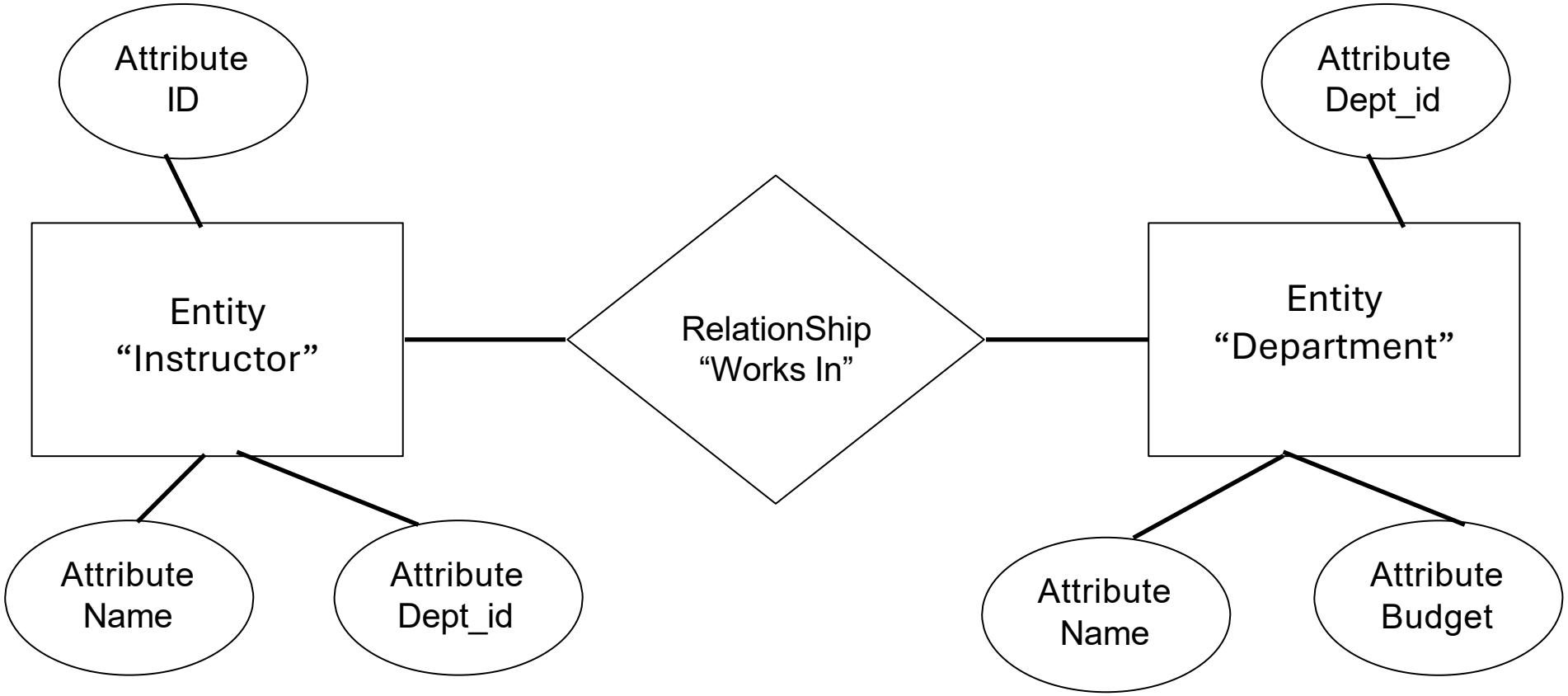
```
UPDATE INSTRUCTOR SET NAME = 'Nina' WHERE ID = 56655;
```

ID	name	dept_id
398227	John	1
48836	Kim	2
29987	Califeri	3
56655	Nina	1

Database Design

- Enforces integrity and relationship in tables through its schema
  - Entity Relationship model
    - Entity Relationship Diagram
    - Like UML Class Diagram
  - Normalization
    - Reduce redundancy
    - Optimize storage and access

Database Design – ERD Representation



Relational Model: Data Models (EERD)

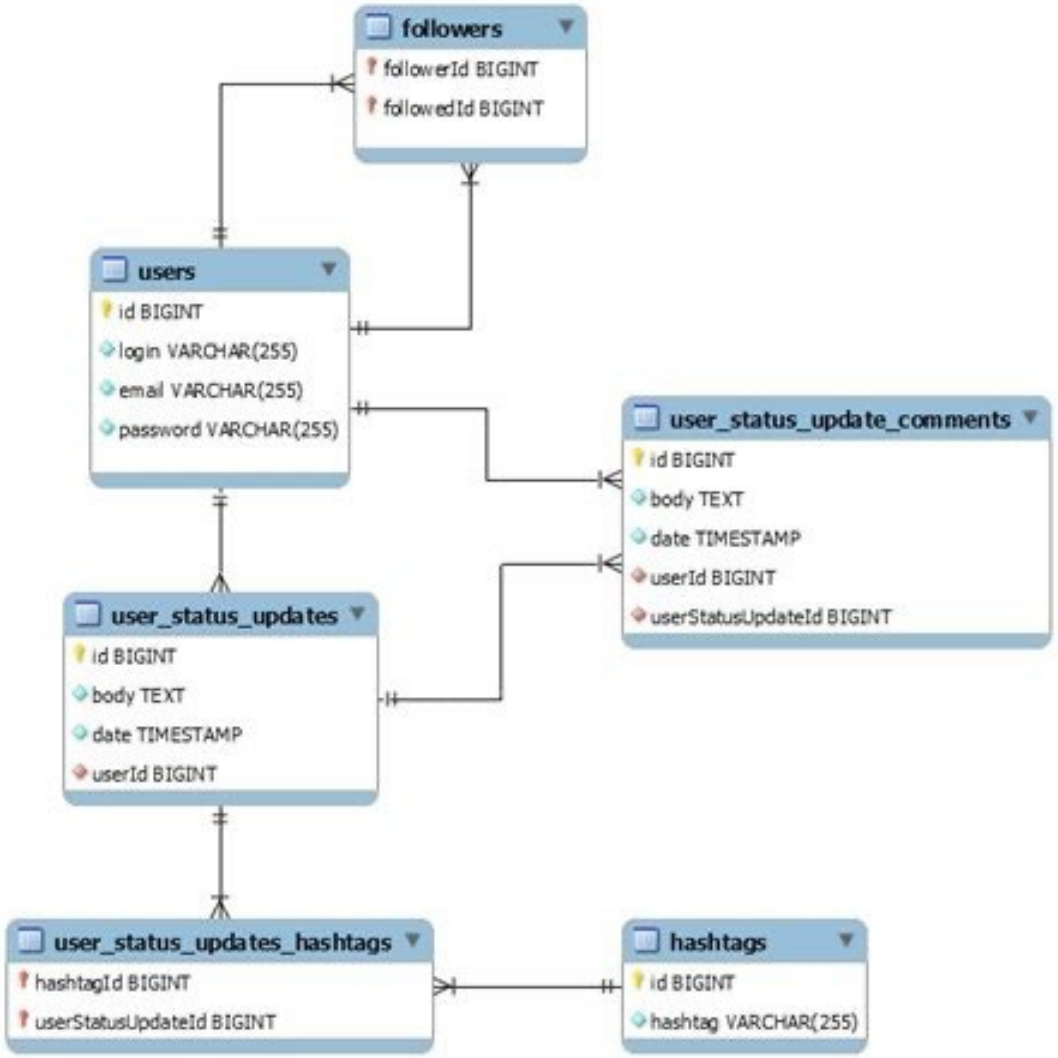


Table of my favorite albums

- Attribute may not contain a set of values or nested record

Album ID	Catalog Number	Album Title	Artist	Released Singles	Label	Publisher Country	Publisher Website
101	PE 34188	Boston - debut	Boston	More Than a Feeling Peace of Mind Long Time	Epic	US	www.epic.com
102	CBS 80959	Born to Run	Bruce Springsteen	Thunder Road Born to Run She's the One	Columbia	US	www.columbia.com
103	PCS-7096	Let It Be	Beatles	Let it Be The Long and Winding Road Get Back	Elektra	US	www.Elektra.com
104	CAQ 33280	That's the Way of the World	Earth, Wind & Fire	Shining Star That's the Way of the World Reasons	Columbia	US	www.columbia.com
105	2GHS 2031	Cats	Andrew	Mr. Mistoffelees	Geffen	UK	www.geffen.com

1st Normal Form (1NF)

- Attribute may not contain a set of values or nested record

Album

Album ID	Catalog Number	Album Title	Artist	Label	Publisher Country	Publisher Website
101	PE 34188	Boston - debut	Boston	Epic	US	www.epic.com
102	CBS 80959	Born to Run	Bruce Springsteen	Columbia	US	www.columbia.com

Single

Album ID	Released Singles
101	More Than a Feeling
101	Peace of Mind
101	Long Time
102	Thunder Road
102	Born to Run
102	She's the One

- Pulled the multi-value attribute, Released Singles, into a separate table

2nd Normal Form (2NF)

- Every non-key column must depend on the whole key, not just part of it

Album

Album ID	Catalog Number	Album Title	Artist	Label	Publisher Country	Publisher Website
101	PE 34188	Boston - debut	Boston	Epic	US	www.epic.com
102	CBS 80959	Born to Run	Bruce Springsteen	Columbia	US	www.columbia.com

Single

Album ID	Released Singles
101	More Than a Feeling
101	Peace of Mind
101	Long Time
102	Thunder Road
102	Born to Run
102	She's the One

- Album ID is the Primary Key
  - Not a composite key, so by definition, it's 2NF

3rd Normal Form (3NF)

- Column values cannot depend on other columns' values in the table

Album

Album ID	Catalog Number	Album Title	Artist	Publisher ID
101	PE 34188	Boston - debut	Boston	601
102	CBS 80959	Born to Run	Bruce Springsteen	602

- Pulled all three Publisher columns into a separate table because the Publisher data depends on the Label value

Single

Single ID	Album ID	Released Singles
201	101	More Than a Feeling
203	101	Peace of Mind
206	101	Long Time
401	102	Thunder Road
408	102	Born to Run
409	102	She's the One

Publisher

Publisher ID	Label	Publisher Country	Publisher Website
601	Epic	US	www.epic.com
602	Columbia	US	www.columbia.com
603	Elektra	US	www.Elektra.com
701	Geffen	UK	www.geffen.com

Database Optimization

- Table design
  - High updates on data should have many tables with few columns
  - High reads on data should have fewer tables with many columns

- Index definitions
  - Balanced Trees (B-trees)
  - Searchable and join columns indexed

Typical use cases for RDBMS

- Structured data sets
  - User information
  - Bank account information

- Need to guarantee data consistency
  - Bank account balance
  - Even after a system crash

- Searches for strings like <searchString> or <searchString%>
  - Wildcards at the end search strings (or none)

- Online Transaction Processing
  - 32K concurrent connections

Tough use cases for RDBMS

- Multimedia
  - Streaming audio or video files
- Genomic data (DNA)
  - 3 Billion base pairs composed of 4 letters (ACTG)
- Searches for strings like %searchString%
  - Wildcards at the beginning of search strings
- Distributed number of high reads
  - Billions of users reading billions of posts (social media)

Java and Relational Databases

- JDBC
  - Java DataBase Connectivity

- Typical JDBC interaction
  - Each row in a table becomes an instance of a class
  - i.e., each row in Instructor table becomes an instance of Instructor.java

- The 3 main objects in JDBC
  - Connection
  - Statement
  - ResultSet

- Object Relational Mapper (ORM)